Final Year Project Proposal

TU858

Phishgaurd

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Table of Contents

Contents

Table of Contents i

Declaration ii

Summary 1

Background (and References) 1

Proposed Approach 2

Deliverables 3

Project Schedule 4

Technical Requirements 4

Conclusion 4

References R1

Appendix A: First Project Review A1

Appendix B: Second Project Review B1

Appendix C: Prompts Used with ChatGPT C1

Appendix D: D1

Declaration

I hereby declare that the work described in this dissertation is, except where otherwise stated, entirely my own work and has not been submitted as an exercise for a degree at this or any other university.

Signed:

*Ismail Khan*_____

Ismail Khan

11/10/2025

Summary

The aim of the project is to develop a web-based application to raise awareness of phishing threats by simulating real-world scenarios in an accessible and engaging format. Scenarios will be crafted with varying degrees of difficulty to challenge both beginners and those with some prior security awareness.

This app is designed with the purpose of accessibility for the necessary education for elders and youth as they are the groups most often the victim of these social engineered attacks. This will be a tailored experience by introducing users with key features such as interactive training modules, simulated phishing examples, immediate feedback for progress tracking and built in accessibility (large fonts, clear icons, simple navigation).

This is done to empower users to recognise and respond to these attacks. The project will use a modern web framework, a database to track user progress and will prioritize strong UI/UX practices. The main motivation for this project being, as phishing continues to be a major security risk as it exploits its victims trust and inexperience it will deliver an accessible, user-friendly web application that guides users in recognising, understanding and safely handling phishing attempts which will contribute to better digital security for all.
As phishing remains a serious digital safety risk that exploits trust and inexperience, this application seeks to promote practical understanding and safer online habits for all.

Background (and References)

Phishing attacks represent a critical cybersecurity threat, with 3.4 billion phishing emails sent daily and 80% of cyber-attacks involving social engineering techniques. These attacks exploit human psychology rather than technical vulnerabilities, making them particularly effective against vulnerable populations including children and elderly users who may lack digital literacy skills. The financial impact is substantial, with phishing-related breaches costing organizations an average of $4.8 million.

Current solutions focus primarily on corporate environments. Commercial platforms like PhishingBox and Microsoft's Attack Simulation Training provide simulated phishing campaigns for employee training. However, these solutions typically target workplace scenarios and lack accessibility features for diverse age groups or cognitive abilities. Previous TU Dublin final year projects have explored related areas, including SecureBuddy (file scanning tools) and network intrusion detection systems, demonstrating student interest in practical security solutions.

Research demonstrates the effectiveness of structured awareness training. A study of Irish SMEs found that cybersecurity workshops reduced phishing click-through rates from 65% to 12%, with participants showing sustained behavioural improvements. Similarly, comprehensive meta-analyses indicate that cybersecurity training produces significant positive effects on user behaviour.

Gamification offers potential for enhanced engagement in educational contexts. Dympna O'Sullivan and colleagues research on gamification ethics in education highlights both benefits (increased motivation, better performance) and considerations (manipulation concerns, declining effectiveness over time).

Their framework emphasizes the importance of ethical design principles when implementing game mechanics in learning environments.
Despite these advances, there remains a significant gap in accessible, age-inclusive phishing awareness tools. MetaCompliance identifies the critical need for security training that accommodates diverse abilities and learning styles. This project addresses this gap by combining proven training methodologies with accessible design principles and ethical gamification approaches.

Proposed Approach

This project will proceed through five phases Research, Requirements gathering, Analysis and Design, implementation, and testing/evaluation to deliver a comprehensive, accessible, and engaging phishing awareness application.

Research:
The research phase will use existing knowledge on phishing education, gamification ethics and accessibility standards. This includes academic studies on training effectiveness(Irish SME training effectiveness), reports from commercial security platforms, Dympna O'Sullivan's ethical gamification framework and past projects. Commercial products such as PhishingBox and Microsoft attack simulation training will be examined to identify best practices and feature gaps. Accessiblity guidelines from WCAG 2.1 will influence UI requirements .

Requirements Gathering:
Requirements will be gathered from user interviews from all age groups. Two primary user groups such are older adult and teenagers who will help understand their digital literacy, risk identification and learning style. Surveys will. Capture their prior experience with online threats and favoured training formats. Accessibility educators will review the draft requirements to ensure best accessibility. Use cases and user stories will then be formalized, documenting scenarios such as identifying suspicious emails and reporting phishing attempts.

Analysis and Design:
In the analysis phase, requirements will be translated into functional and nonfunctional specifications. A modular architecture will be defined, comprising a responsive front end, a gamified scenario engine, and a backend data service. Interaction flows will map user journeys through adaptive phishing simulations, immediate feedback, and progress tracking. Wireframes and high-fidelity mockups will be created to validate navigation, colour contrast, and assistive technology compatibility. Ethical design checkpoints will be

embedded at each stage to prevent undue manipulation, drawing on O'Sullivan's checksheet.

Implementation:
The application will be built on a modern web stack. On the front end, React will be used with ARIA roles so that screen readers and keyboard navigation work seamlessly. The back end will be powered by Django REST framework, which will handle user login, scenario rules, and storing data in PostgreSQL. Gamification will be added through modules for points, badges and adjusting difficulty based on performance. Feedback and analytics will update in real time using WebSockets. We will use continuous integration to keep code quality high and run security scans, and Docker will package the app for easy deployment.

Testing/ Evaluation:
Testing and evaluation will combine automated checks with real world trials with actual users. First, automated tests will confirm that the interface meets WCAG 2.1 guidelines and that all features function correctly. Next, we will invite older adults and teenagers to use the application while we observe how easily they complete tasks, the number of errors they make and their overall satisfaction.

We will assess participants' ability to identify phishing attempts before and after the training to measure improvement. Engagement data such as time spent on each scenario, completed challenges and retained badges will be collected to see how the gamification elements affect user behavior. Insights from both quantitative metrics and participant feedback will guide iterative refinements until the application effectively teaches users to recognise and report phishing while remaining accessible and engaging.

Deliverables

First, I will put together a clear write-up of everything I learn about phishing training, gamification, and accessibility, so we have one place to look for design ideas and supporting documentation in the project report. Next, I'll turn our interviews with older adults and teens into a requirements document full of real user stories, use cases and accessibility checks. Then I'll sketch out and refine the app's layout and data structure with wireframes, mockups and an ER diagram for users, scenarios, and feedback.

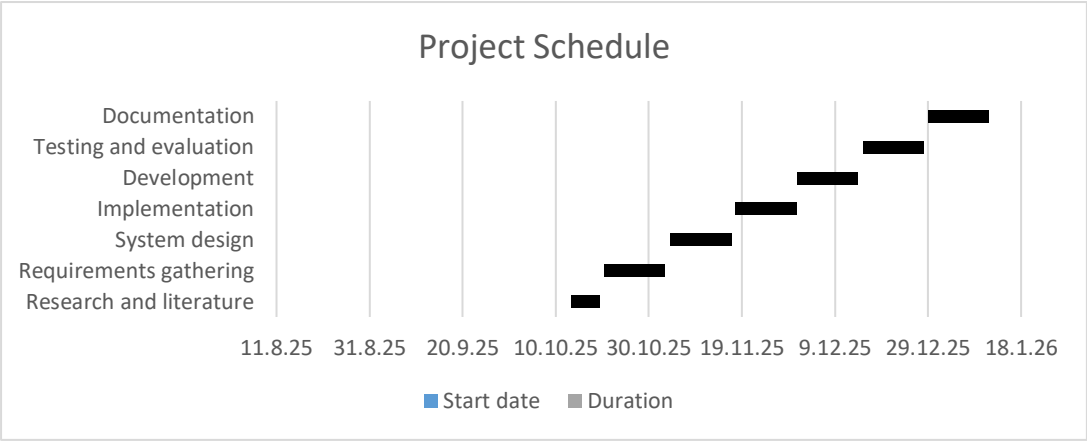
Once the design is solid, I'll build the actual app using React on the front end, Django REST on the back end and PostgreSQL for storage, complete with points, badges, adaptive difficulty, and real-time feedback. I'll also write automated tests for both functionality and WCAG compliance and set up a CI pipeline plus Docker containers so the app is easy to deploy.

Finally, I'll run a mix of lab tests and live user sessions to see how well people learn, how usable the interface is and whether it meets accessibility standards, then collect that data and feedback in a final evaluation report with recommendations for any last tweaks.

Project Schedule

I'll spend the first few months digging into existing research and talking with potential users to nail down what matters most. After that I'll sketch out design ideas and data structures before moving on to actually building the core features of the app. Once the main functionality is in place I'll run automated tests and gather feedback from real users to iron out any kinks. Finally I'll wrap everything up with a thorough evaluation report and make sure all the documentation is ready for submission.

- Gantt Chart required



Technical Requirements

I will need a capable development machine, either a Mac or PC that can run Docker containers and host our local servers, plus a mix of browsers on desktop and mobile and at least one screen reader for accessibility checks. On the software side we will install Node.js and npm to build the React interface, Python 3 with Django REST framework for the API, and PostgreSQL to store user and scenario data.
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WebSockets support via Django Channels or a similar library will power real time feedback. We will use Git with a CI tool such as GitHub Actions to catch errors early and Docker to package everything for easy deployment. In the cloud we will spin up our containers on AWS or another provider, store videos and static files in S3 or equivalent object storage, and secure the site with SSL/TLS certificates.

We will also integrate logging and monitoring using the ELK stack, Datadog or a comparable service to keep an eye on performance and errors, and we will run automated accessibility tests with tools like axe-core to ensure WCAG compliance.

Conclusion

This project will deliver an accessible, engaging phishing awareness application that fills a crucial gap in current training tools. By combining thorough research on phishing education, gamification ethics and accessibility standards with real user insights from older adults and teenagers, we will ensure the application meets diverse learner needs.

A clear requirements document will guide us as we design intuitive wireframes, mockups and data models. Building the app on a modern web stack with React and Django REST, supported by PostgreSQL and real-time feedback via WebSockets, will allow for flexible features like adaptive difficulty and live analytics. Automated testing and continuous integration will keep the code reliable and WCAG-compliant while Docker makes deployment straightforward.

Finally, lab tests and live user sessions will verify that the app effectively teaches users to spot phishing, remains easy to use and upholds accessibility standards. The final evaluation report will capture lessons learned and recommend refinements, setting the stage for a polished tool that helps protect everyone from phishing threats.

References

Hint:

Use Zotero to manage your references (see Brightspace resources).

Use the **Harvard** or **IEEE** numbering referencing style

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Appendix A: First Project Review

Hint: review a past project from the library website that relates to your project idea.

Title: A Computer Application for Teaching Children Computer Skills

Student: Jennifer Seery

Description (brief): The report presents a web application that delivers a sequence of lessons to teach foundational computer skills to children. Beginning with basic mouse and keyboard exercises, the curriculum advances through tasks such as creating documents and navigating simple web pages. The user interface employs bright colours, intuitive icons and concise instructions, and each completed activity triggers immediate visual or auditory confirmation to reinforce learning.

What is complex in this project: the most complex aspect of this project is finding the right balance between providing clear guidance and allowing children the freedom to explore. Another complex aspect being ensuring each lesson was simple enough to follow without causing frustration, while also building in flexibility and managing multiple user sessions so that each child's progress was reliably saved.

What technical architecture was used: Technically, the application relies on an ASP.NET server framework for lesson logic and user authentication. A MySQL database maintains records of lesson completions, quiz scores and user preferences. The front end uses HTML5, CSS3 and JavaScript to ensure responsiveness and compatibility across desktop and tablet devices. Teacher dashboards query the database through RESTful endpoints to display progress reports and highlight areas needing attention.

A significant strength of the project is its clear lesson scaffolding, which guides learners through increasingly complex tasks without overwhelming them. The instant feedback mechanism enhances engagement and supports mastery of each skill. However, the application does not offer customisation for different learning styles or special educational needs. Its monolithic architecture may hinder future feature extensions and performance optimisations. Additionally, accessibility features for users with visual or cognitive impairments are limited, reducing the tool's inclusivity.

Appendix B: Second Project Review

Hint: review a past project from the library website that relates to your project idea.

Title: FitnessEnhanced

Student: Ahmed Alazawy

Description (brief): FitnessEnhanced combines web and mobile interfaces to guide users in setting personalized exercise goals, logging workouts and receiving customized coaching tips. The application offers a dashboard that displays progress charts, a library of on-demand exercise videos and automated reminders designed to reinforce healthy habits.

What is complex in this project: Generating real-time coaching recommendations is particularly challenging because the system must analyse each user's performance history, identify patterns in workout data and adjust advice accordingly. Coordinating seamless video streaming, analytics and push notifications across different devices while preserving a responsive user experience adds another layer of technical intricacy.

What technical architecture was used: The front end of FitnessEnhanced is built with React to serve both web browsers and hybrid mobile applications. Behind the scenes, a Node.js and Express API layer manages user authentication, workout data processing and recommendation logic. User profiles, workout records and video metadata reside in a MongoDB database. Exercise videos are stored on AWS S3 and a serverless function processes incoming workout logs to update performance metrics in near real time

Explain key strengths and weaknesses of this project, as you see it. The personalized coaching engine stands out as a major strength because it adapts recommendations continuously as new workout data arrives. The choice of a NoSQL data store offers flexibility in handling diverse exercise types and user-generated content. On the other hand, reliance on external video streaming services can lead to unpredictable latency under heavy load and the monolithic design of the API layer may pose challenges when scaling individual features. Finally, although reminders help maintain user engagement, they must be carefully tuned to avoid overwhelming users and causing notification fatigue.

Appendix C: Prompts Used with ChatGPT

- “Summarize global phishing attack trends and key statistics from reputable industry reports published between 2020 and 2025.”
- “List major social engineering tactics used in email-based phishing campaigns and their psychological impact on users.”
- “Compare the effectiveness of phishing simulation platforms designed for corporate training versus educational settings.”
- “Identify best practices for integrating gamification ethically into cybersecurity awareness programs.”
- “Extract WCAG 2.1 success criteria most relevant to interactive web applications for users with cognitive impairments.”
- “Outline architectural patterns for real-time feedback systems in web applications using WebSockets.”
- “Describe accessibility testing tools and methodologies appropriate for evaluating screen reader compatibility.”
- “Review case studies on phishing awareness interventions in small and medium enterprises, including outcome metrics.”
- “Highlight common challenges and solutions in deploying containerized web applications with Docker and CI/CD pipelines.”
- “Summarize AWS services suitable for hosting static video content and supporting API back ends for e-learning applications.”

Appendix D: